

Max Johnson

(559) 765-5207 | Fresno, CA | maxjohnson87@protonmail.com | max112358.github.io

SKILLS

- **Languages & Frameworks:** C++, C#, Python, JavaScript, SQL, Lua, HTML/CSS
- **Web & Cloud Software:** Full-Stack Architecture, RESTful APIs, Node.js, PostgreSQL, Cloudflare, AWS/R2, Render
- **Core Computer Science:** Data Structures & Algorithms, Systems Programming, Memory Management, State Machines
- **Developer Tools:** Git/GitHub, Linux/Unix, CI/CD Pipelines, Debugging Tools, Unit Testing & Automation

PROJECTS

Distributed Cloud Infrastructure & Full-Stack Platform

conniption.xyz

- Architected and deployed a multi-tier web application using Cloudflare for edge routing, Node.js/Render for backend microservices, and PostgreSQL via Aiven
- Integrated Cloudflare R2 object storage to streamline asynchronous file uploads, implementing multi-threaded serialization to reduce latency for high-concurrency requests
- Configured RESTful API endpoints and client-side dynamic rendering for real-time thread updating and media delivery

Barotrauma Circuit Blueprint Interconnect (Mod)

[https://steamcommunity.com/sharedfiles/filedetails/?id=](https://steamcommunity.com/sharedfiles/filedetails/?id=3301014847)

[3301014847](https://steamcommunity.com/sharedfiles/filedetails/?id=3301014847)

- Engineered a C#-to-Lua translation layer to dynamically parse, serialize, and manipulate complex in-game digital logic circuit node topologies
- Built custom state-saving and loading algorithms serving over 3,000 active users, ensuring persistent data storage across live gameplay sessions

Deep Learning Age Estimation Model

<https://github.com/Max112358/Facial-Recognition-of-Age>

- Developed and trained a convolutional neural network (CNN) using Python and TensorFlow/Keras on the IMDB-WIKI dataset to predict age ranges from image data
- Applied data preprocessing, feature scaling, and hyperparameter tuning to optimize prediction accuracy and model convergence

Tabletop Dice Roller

https://max112358.github.io/tabletop_dice_roller/

- Built a custom browser-based tabletop dice engine in vanilla JavaScript that parses complex RPG formulas (advantage/disadvantage, dice pools, exploding/reroll dice, recursive variables, and success checks) and renders results to a clipboard-integrated UI.
- Designed a modular single-page app with world/character profile management, persistent localStorage state, drag-and-drop reordering, modal CRUD flows, and JSON import/export for tabletop character sheets.

Bottle Maker

https://max112358.github.io/bottle_maker/

- Built a browser-based procedural 3D bottle designer with Three.js, featuring real-time parametric mesh generation, physically based glass/liquid/cork materials, procedural canvas-generated textures, and dynamic paper labels.
- Implemented GLB model export plus JSON save/load for configurations, with a responsive UI that live-updates geometry, estimates triangle counts, and supports customizable dimensions, fill levels, and label text/fonts.

WORK EXPERIENCE

Prop & Automation Software Technician

Aug 2025 — Present

Fresno Fright Nights

Fresno, CA

- Programmed and deployed embedded C/C++ firmware across ATtiny85 and Arduino controllers to govern automated state transitions and real-time execution loops
- Developed automated control scripts to synchronize event triggers, audio streams, and low-latency digital outputs during live operations
- Performed system-level debugging, sensor signal diagnostics, and dynamic fault recovery to guarantee 100% operational reliability

Computer Science Tutor

Aug 2022 — Dec 2024

Fresno City College

Fresno, CA

- Instructed computer science students in C++, Python, object-oriented design, and foundational Data Structures & Algorithms
- Conducted regular code reviews with students, focusing on memory management, pointer manipulation, time complexity analysis, and clean code practices
- Guided students through low-level debugging techniques using GDB and IDE diagnostic tools to identify and resolve segmentation faults and memory leaks

Assembly & Automation Technician

Feb 2020 — Aug 2022

Maxco Supply

Parlier, CA

- Programmed and debugged industrial PLC logic scripts to control sequence-of-operations for automated packaging machinery
- Diagnosed complex electrical and logical system failures using schematics, multimeters, and systematic state-testing procedures

EDUCATION

Fresno State University

Fresno, CA

Bachelor of Science in Computer Science

Graduated Dec 2024

- Relevant Coursework: Algorithms & Data Structures, Systems Programming, Simulation, Artificial Intelligence

Clovis Community College

Clovis, CA

Associate of Science in Mechatronics

Aug 2019 — Aug 2021

- Relevant Coursework: Industrial Automation, Circuit Analysis, PLC Programming Logic